

Metadata Collection for MAAP: A Metadata-Aware Application Profiling Framework

Shannon Crandley
Computer Engineering
Tufts University

Abstract—Memory systems have been improving since they were first created, but recently, the improvements towards performance are outgrowing the improvements of memory latency. MemSysExplorer is an interface that uses memory profiling systems to output workloads of memory characteristics that can be used to create one data set of all the different memory specifications. This portion of MemSysExplorer follows the collection of the metadata through the applications team. Execution times from SPEC2017 workloads that are found and analyzed to contribute to the metadata set will later be published.

I. INTRODUCTION

Modern computer architecture is constantly growing and improving, so it is crucial that memory performance improves with architecture. Workloads are becoming more complex, which means memory systems must meet the performance requirements needed for these workloads. It can be difficult for hardware architecture engineers to understand where to improve memory systems and how their chip compares to other chips currently available. Right now, there are no profiling systems that output all the needed memory statistics of a memory chip that make memory characteristics readily available and easily accessible. The future of memory systems is improved bandwidth, decreased latency, and energy efficiency, but finding these data sets for memory chips can be difficult. Existing profilers only focus on certain metrics that do not produce metadata needed to understand new, complex workloads.

This paper presents initial work towards creating the MAAP, a Metadata-Aware Application Profiling, data set that will be published in the future. Thus, MAAP addresses these issues and outputs a complete and compatible data set that researchers and engineers can access and use for improving memory systems. The dataset integrates data across multiple profilers, such as DynamoRIO, NVBit, perf, Nsight Compute, and Sniper, so data is readily accessible in one place. This semester, this research consisted of learning what memory profiling is, how it applies to the project, and how to use profilers with SPEC CPU2017 benchmarks. SPEC CPU2017 benchmarks provide adoption and ability that represent diverse real-world workloads, allowing for collection of reproducible and comparable memory statistics. In this study, DynamoRIO, a dynamic binary instrumentation that observes and manipulates every instruction at runtime in a program without requiring access to the program's source code, is used on SPEC CPU2017 benchmarks. The insights gained from this work inform how to

approach additional profiling tools in future phases of MAAP, ultimately contributing to the final metadata set.

II. MOTIVATION AND GOALS

The motivation for this project arises from the lack of easily accessible and readily available metadata showing the behavior of modern memory systems, as well as the absence of a profiling tool that can create a consolidated, comprehensible, and compatible data set of memory statistics of workloads. Workloads are constantly changing and becoming more complex, so profilers are used to understand their compilation, performance, and statistics, which is necessary for understanding how compatible the memory system is for the program. As applications continue to evolve in complexity and scale, understanding how workloads interact with memory systems has become increasingly important. MAAP addresses the demand for profiling, as it is essential for evaluating compilation behavior, runtime performance, and memory access characteristics.

As complexity of workloads increases, the gap between memory and processors widens, leading to an increase in memory latency, energy consumed from memory architecture, and bandwidth bottlenecks. Due to these reasons, the need for memory system optimization becomes more important. With MAAP, hardware architects and researchers are able to identify performance bottlenecks and evaluate memory architecture to improve memory characteristics. This research is meant to expose crucial memory metadata for the development of future memory system designs.

The goals of this project is learn about how workloads perform on different memory systems and how this can be improved with new memory systems. This project aims to contribute to the future of memory systems, which will be fast, reliable, and extremely carbon efficient. The creation of the data set for MAAP is a the first step for researchers to find how workloads perform with certain chip specifications, Researchers can use this dataset to identify performance trends, evaluate architectural trade-offs, and create the design of next-generation memory systems.

Personal goals of this project include learning more about memory systems, learning how to use profilers, and understanding benchmarks used for CPU performance. The future of memory is ever-changing, and the importance of new architecture is rapidly increasing, as energy efficiency can affect climate change, and accessibility for memory architecture and statics grows.

Benchmarks	Profiler	Experiment Type	Memory Statistic	Trace	Samples	Metadata	Baseline	Progression	Note
SPEC2017	Dynamorio	Execution Context, Sample Analysis	Shannon	Not Started	Duc	Completed	Completed	25%	Duc, Shannon
	Sniper	Hardware Topology, Sample Analysis	In Progress			In Progress	Completed	25%	Shannon
	Perf	Execution Context, Sample Analysis	Gloria				Completed	25%	
PolybenchGPU	NCU	Hardware Topology, Sample Analysis					Not Started	30%	
	NVBIT	Execution Context, Sample Analysis					Not Started	30%	
MLPerf	NCU	Hardware Topology, Sample Analysis						0%	
	NVBIT	Execution Context, Sample Analysis						0%	
	Sniper	Hardware Topology, Sample Analysis						0%	
Cloud Computing								0%	
EMBBench	Sniper	Hardware Topology, Sample Analysis						0%	

Fig. 1: MemSysExplorer Data Sheet Organizer

III. EXPERIMENTAL DESIGN AND TOOLS

A. Experimental Environment

All experiments were conducted on the Kirin server, which was the primary remote execution environment for benchmark runs and profiling experiments. The initial setup began with entering the Kirin server, navigating the remote directory, and creating configuration files with the SPEC CPU2017 monitor wrapper to understand system behavior. The remote file system was initially confusing, and sometimes when running the benchmarks, results were written to the wrong directories. Thus, the configuration files were updated to ensure that all outputs were written to the correct directories.

B. Benchmark Suites

The primary workloads used in this research was SPEC CPU2017, which measure and compare different memory statistics of processor, memory, and compiler performance. Initial dry runs were performed to validate proper benchmark execution, understand how to execute these benchmarks, and to establish baseline performance statistics.

In addition to SPEC CPU2017, the Alberta workloads, a different set of benchmarks, were explored towards the end of the study. These were used to diversify the data set for MAAP and extract more data characteristics with different workloads. However, full integration of these workloads was not completed within the scope of this study.

C. Profiling Tools

Dynamic Binary Instrumentation tools were used to collect memory metadata so access to the workloads source code was unneeded. The primary instrumentation tool used was DynamoRIO. DynamoRIO was executed through the MemSysExplorer framework within the server environment DynamoRIO, allowing runs of DynamoRIO on the SPEC CPU2017 workloads.

Additionally, the Sniper simulator was utilized for the Alberta workloads. The Sniper simulator analyzes power consumption and performance on different workloads. The Sniper

SPEC2017 Baseline			
Suites	Benchmarks	Est. Base Run Time	Est. Base Rate
Integer Rate	500	286.348	5.560
	502	222.939	6.352
	505	345.425	4.678
	520	-	-
	523	277.270	3.809
	525	-	-
	531	266.063	4.307
	541	438.068	3.780
	548	267.574	9.792
	557	331.575	3.257

Fig. 2: Example Data Sheet of Benchmark Data

simulation on Alberta workloads will be included in the MAAP data set, and the execution of these simulations will happen later.

D. Data Collection

Data collection began with dry runs of SPEC CPU2017 using their executable, the `runcpu` command, which creates a non-profiled baseline of memory statistics. This verifies stability of the memory, as well as validity of the memory statics and workloads. After the dry runs were completed, DynamoRIO was used to profile the SPEC CPU2017 workloads. The tool collected execution times and memory statistics of the workloads. Unfortunately, many runs failed, especially the integer speed suite, which needed debugging. Due to this setback, DynamoRIO was not executed on the benchmarks as many times as desired.

For the Alberta workloads, only a couple benchmarks were able to be executed under the Sniper profiler, but not enough to create a valid data set.

E. Data Management

To organize data collection of the different profilers and members of the research team, a master data sheet, shown

SPEC CPU2017: Baseline vs DynamoRIO

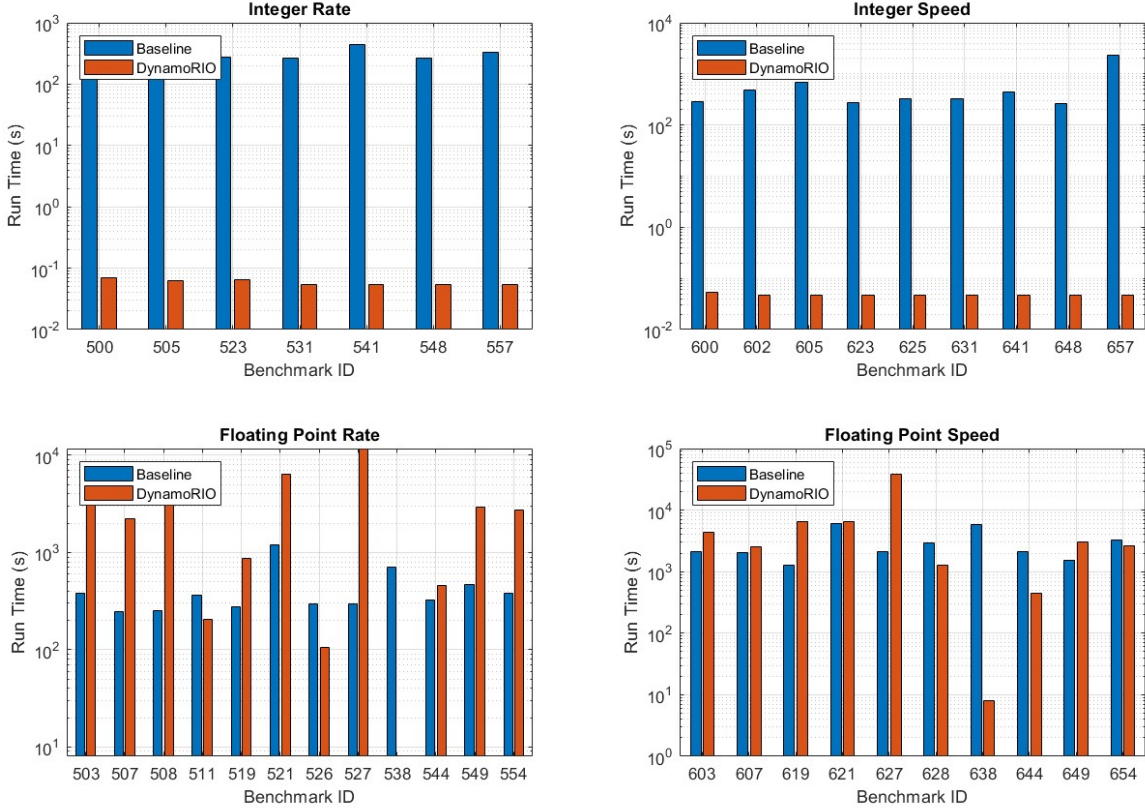


Fig. 3: SPEC CPU2017 Base Run Times on the Different Benchmarks

in Figure 1, was created in Google sheets to consolidate information and tasks of the team. The sheet consisted of the benchmarks, the profilers, the data from the profiler, and the progress of the profilers. For some of the progress tabs, they were replaced names of the team member performing the tasks. Additional subsheets, as shown in Figure 2, were created to document data from the executions run of the profilers. Thus, all the data and progress was in one place.

IV. RESULTS

Figure 3 illustrates the performance comparison between the Baseline and the DynamoRIO run times across SPEC CPU2017 benchmarks, showing results of both Integer and Floating Point results. The figure is organized into four subplots showing depending on the suite of benchmarks that were executed, with all run times plotted on logarithmic scales due to the large difference in run times across the benchmarks.

A. Integer Benchmarks

For the integer workloads, the DynamoRIO outputs are very different than the baselines. In the Integer benchmarks (Figure 3, top), Baseline runtimes are relatively regular, whereas DynamoRIO runtimes are all under 1 second, meaning that

there is a possibility that DynamoRIO outputted the average instruction time and not the runtime of the program. The Integer Rate benchmarks all have very similar outputs of time for both baseline and DynamoRIO. Similarly, the Integer Speed benchmarks (Figure 3, top right) demonstrate a consistent time for each workload, but the reason for these numbers might be similar to the Integer Rate, where instruction time was outputted instead of execution time.

B. Floating Point Benchmarks

The Floating Point benchmarks are more reliable than the Integer benchmarks, as the run time of DynamoRIO is relative to the baseline execution times. DynamoRIO outputs on the benchmarks is much greater than the baseline statistics for Floating Point Rate, but there is still some variation for whether the execution time is greater than or less than the baseline. DynamoRIO did not output anything for benchmark 538. There are two benchmarks with DynamoRIO run times that are less than the baseline, which are 511 and 526. As for the floating point speed, there are more DynamoRIO executions that are less than the baseline, which are 628, 638, 644, and 654. Additionally, for Floating Point Speed, the DynamoRIO run times are much more similar to the baseline run times. This

variability indicates that floating point workloads are more sensitive to the specific types of instructions in the benchmarks, and that DynamoRIO interacts more consistently with these workloads.

C. Analysis

Overall, the results show that the overhead introduced by DynamoRIO is highly workload-dependent. The Integer workloads seem to have large error with the DynamoRIO run times, meaning that these will need to be rerun and debugged. There is a chance that the run times measured were average instruction execution time, as that is what DynamoRIO measures, but there was probably an error with the files that caused the outputs to be incorrect. These workloads show a large range of different outputs across the workloads, making it difficult to come to a conclusion about the data. The disparity between rate and speed measurements further emphasizes that benchmark execution, whether focused on repeated iterations or full completion time, can influence the observed performance. These findings are important for researchers and hardware architects who rely on dynamic instrumentation for profiling. Careful benchmark selection and interpretation of results are necessary, particularly for integer benchmarks which had much error and disparity across run times.

D. Summary

In summary, there seems to be error across DynamoRIO's run times on these benchmarks, especially for Integer Speed and Integer Rate. As for the Floating Point benchmarks, DynamoRIO runs much more consistently, allowing easier analysis for floating point-heavy applications. The inconsistent values mean that although there is probably an error in the execution of DynamoRIO, it can teach that data may not always be consistent and data analysis is vital, especially upon creating reliable and realistic data sets that are available for public use and research. DynamoRIO will need to be run again for the metadata dataset for MAAP, which is necessary, as these run times need to be as consistent as possible.

V. FUTURE WORK

The next step is to expand on the data collected for MAAP and continue contributing to development of the metadata dataset. This will involve executing different workloads across many profiling tools to get a broader dataset with more memory and performance statistics. This dataset will have real-world applications that can be used by researchers and engineers alike.

An objective of future work is to standardize the structure of the collected metadata to make it as easily readable as possible. Multiple trials of the profilers on the workloads will be run to ensure reproducibility, making the dataset more reliable and valid.

Once completed, the MAAP dataset will be made available to the public, therefore researches and engineers can develop future memory systems. By providing an a rich data-set of metadata, MAAP aims to support research in performance

optimization, energy-efficiency, and complex workload-aware memory system development.